

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Electrical Engineering Semester : Fifth Subject Code & Name: BTEE505 Illumination Engineering Max Marks: 60 Date:19/01/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Three of the following.		12
A)	Explain the hazards caused due to optical radiation	CO1	04
B)	Compare artificial light and natural light	CO1	04
C)	State the advantages of good lighting	CO1	04
D)	Explain briefly “polar photometer”	CO1	04
Q.2	Solve Any Two of the following.		12
A)	State any six characteristics of gas discharge lamps	CO2	06
B)	Explain drawing a schematic diagram a gas discharge phenomenon	CO2	06
C)	Give a list of various types of lamps used in lighting	CO2	06
Q. 3	Solve Any Two of the following.		12
A)	Explain following reflections (i) Specular (ii) Diffuse (iii) mixer	CO3	06
B)	What is reflector? State necessity and also state their advantages	CO3	06
C)	Draw and explain instant star circuit for fluorescent tube	CO3	06
Q.4	Solve Any Two of the following.		12
A)	State important objective of interior design and explain in brief	CO4	06
B)	What lighting sources and accessories ,fitting luminaries required in residential lighting	CO4	06
C)	What are the different types of hotels and different areas to be illuminated in hotels?	CO4	06
Q. 5	Solve Any Two of the following.		12
A)	Explain the important points in street lighting	CO5	06
B)	State the objective of road lighting	CO5	06
C)	What are the two types of fiber optic cable	CO6	06